

EXAM 7 – SPRING 2015

21. (1 point)

a. (0.5 point)

Briefly describe two benefits of using copulas to express correlation from joint loss distributions.

b. (0.5 point)

Identify a copula that is more appropriate than the normal copula for modeling insurance loss at the portfolio level, and briefly describe a feature of this copula that makes it more appropriate than the normal distribution for modeling insurance loss data.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 21	
TOTAL POINT VALUE: 1	LEARNING OBJECTIVE: C3: Evaluate and select appropriate models to handle diverse risks, including stochastic approaches.
SAMPLE ANSWERS (BY PART, AS APPLICABLE)	
Part a: 0.5 point	
<i>Sample Answer 1</i> • A copula can join any distributions, regardless of what family they are from • A copula can reflect increased correlation between the distribution in the tail <i>Sample Answer 2</i> • Copulas allow for recognition of varying correlations at different levels of a distribution. This helps for modelling risks that aren't always correlated, but are in the tail. • Copulas also facilitate simulations of events, which can help in understanding how to mitigate risks. <i>Sample Answer 3:</i> • Forces correlations between marginal distributions without making any assumptions regarding causality. • Provides significant flexibility in quantifying tail exposure, which is an effective ERM approach. <i>Other accepted responses – any two of the following received full credit</i> • Copulas can join any two distributions and show this without needing to disclose the underlying distributions (so can keep proprietary info safe but still show correlation) Quantify correlation along entire distribution • Copulas can be used to report dependencies to external parties (maybe a rating agency) without giving away the underlying distribution of losses) • It allows for the creation of graphs using various statistics along the distribution • As opposed to scatterplots, provides a measure of the correlation between the joined variables. • Can select a copula based on the correlation want to use. Heavy in the right tail, just use a copula with heavy dependencies in the right tail, etc. • Allows for nonlinear correlations. For example, in insurance some lines may be loosely correlated at small percentiles, while being heavily correlated in the tail.	
Part b: 0.5 point	
<i>Sample Answer 1</i> I would use the Heavy Right Tail Copula (HRT). This copula has a light left tail and a heavy right tail, so it will reflect the increase in correlation of insurance losses in the righty tail (during extreme events). <i>Sample Answer 2</i> We could use a Gumbel copula since it has a heavier right tail than the Normal copula, this can better reflect the skewness of insurance losses.	

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EXAMINERS' REPORT
Most candidates performed very well on this question, as it tested basic concepts regarding copulas.
Part a
The vast majority of candidates received full credit; almost all candidates received at least some credit. In order to receive full credit candidates needed to provide two distinct benefits of using copulas. Some candidates did not receive full credit when the second benefit they listed was essentially the same as the first benefit.
Part b
Over 95% of candidates received full credit on this part. The candidate needed to recall one copula that is more appropriate for modeling insurance than the Normal copula and briefly describe why it is better. The most common error for this problem was candidates who mistakenly selected Frank's Copula as having a fatter tail than the Normal, which is incorrect. Other candidates only named a copula and failed to describe a feature of the chosen copula that would make it more appropriate than a Normal one.